

Using Java 8 Streams

Programming Projects:

1) A Student Stream

- Start Eclipse and create a new Java project called 'lab8'
- Download the associated **Student.java** and **Driver.java** files (also shown below)
- Add those two files to the project you have created, and ensure you can execute the driver class. (output should be shown in the console).
- Using the Streams API complete the following tasks within the main() method of the Driver class -
 1. Create a Stream from the students List, which uses the forEach() terminal method to print information about each student to the console. Note: the Student class has a toString() method so can be passed directly to the System.out.println method.
 2. Create a Stream from the student list which filters the stream to only students which have a grade of over 70%, then uses the forEach() terminal method to print information about each remaining student.
 3. Create a Stream from the student list which uses map() to change each Student object into a student name (String) value. The name should always be uppercase, then sorted alphabetically. Collect the results into an array called 'studentArray' then print the array content using a command such as -

```
Object [] studentArray = ...stream methods  
System.out.println(Arrays.toString(studentArray));
```

4. Create a Stream from the student list which uses mapToDouble() to create a specialized DoubleStream. Then call a method on that DoubleStream to calculate the average grade for all students. Finally call `getAsDouble()` to access and store the result in a variable called 'aveGrade' and print to the console.

2) Generated Streams

- Using the Streams API complete the following tasks within the main() method of the Driver class -
1. Use the `IntStream.range()` method to create a stream of integers from 90 to 100. Map each element of the stream into the cubed version of the number then output to the console using a call to the `forEach()` terminal method.
 2. Use the `Stream.generate()` method to create an infinite stream of random numbers, `map()` these to the range 0 to 5 and ensure they are integers. *Hint*: use a Lambda expression such as -

```
n -> (int)(n * 5 + 1)
```

Now use the `limit()` intermediate function to ensure only 10 values are generated. Collect the values into a List of integer values, then display the list using a traditional for loop, e.g.

```
List<Integer> numList = ...stream methods
```

```
for(int next : numList)
    System.out.println("Next value is : " + next);
```

3. What additional intermediate method could be added to the above statement to ensure any duplicate values were removed from the resultant collection? Add this to the solution.

```

/**
 * A student class for storing overall grades.
 */
public class Student {

    /**
     * The student ID
     */
    private int id;

    /**
     * The student's name
     */
    private String name;

    /**
     * The student's overall grade.
     */
    private float grade;

    /**
     * @return the student ID
     */
    public int getID() {
        return id;
    }

    /**
     * @return the student's name
     */
    public String getName() {
        return name;
    }

    /**
     * @return the student's overall grade.
     */
    public float getGrade() {
        return grade;
    }

    public String toString() {
        return id + " Name : '" + name + "' Grade: " + grade;
    }

    /**
     * Constructor
     * @param id
     * @param name
     * @param grade
     */
    public Student(int id, String name, float grade) {
        this.id = id;
        this.name = name;
        this.grade = grade;
    }
}

```

```
import java.util.ArrayList;
import java.util.List;

public class Driver {

    public static void main(String[] args) {

        List<Student> students = new ArrayList<Student>();

        students.add(new Student(1, "JJ. Jognson", 38));
        students.add(new Student(2, "P. Tibbs", 48));
        students.add(new Student(3, "W. Major", 28));
        students.add(new Student(4, "P. Turner", 60));
        students.add(new Student(5, "A. Ball", 20));
        students.add(new Student(6, "B. Smithy", 90));
        students.add(new Student(7, "R. Jules", 71));
        students.add(new Student(8, "T. Verne", 97));

        // create your Stream solutions here

    }
}
```